

Progress Report

Day 17 — Dashboard and Teams Module Integration

Task Management MERN Stack Platform

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1. Day-17 Objective

The objective of Day 17 was to integrate backend team and task data into the frontend application, develop the dashboard overview page, implement team management views, and establish loading and error handling for asynchronous data retrieval.

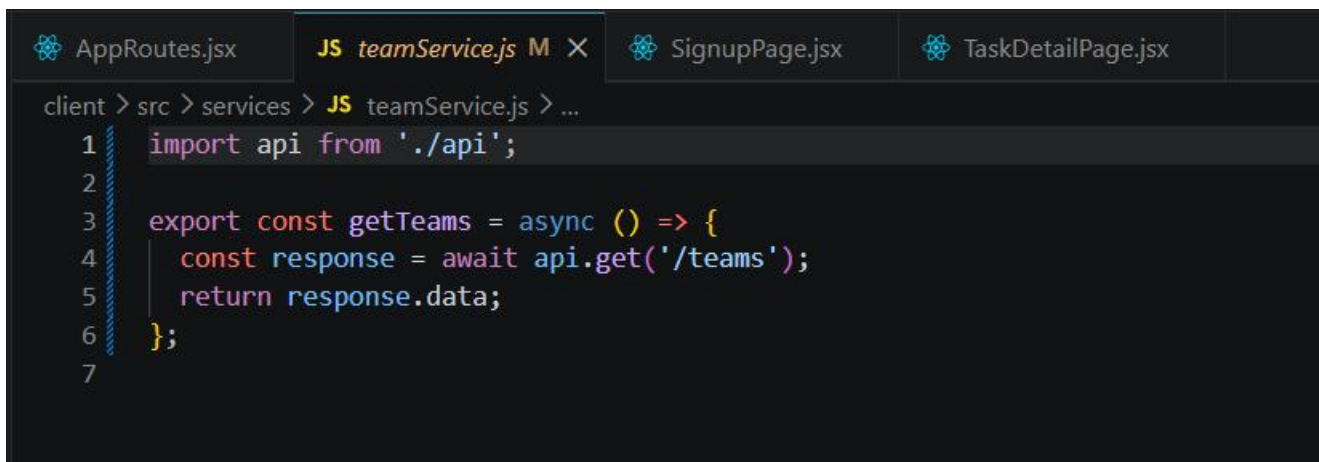
2. Tasks Completed

Team Service Implementation

- Implemented frontend service methods for retrieving team data from backend APIs.

Configured functionality:

- Team retrieval through API requests
 - Centralized service layer integration
 - Reusable data access methods
- Established communication between frontend components and team management endpoints.



```
client > src > services > JS teamService.js > ...
1  import api from './api';
2
3  export const getTeams = async () => {
4    const response = await api.get('/teams');
5    return response.data;
6  };
7
```

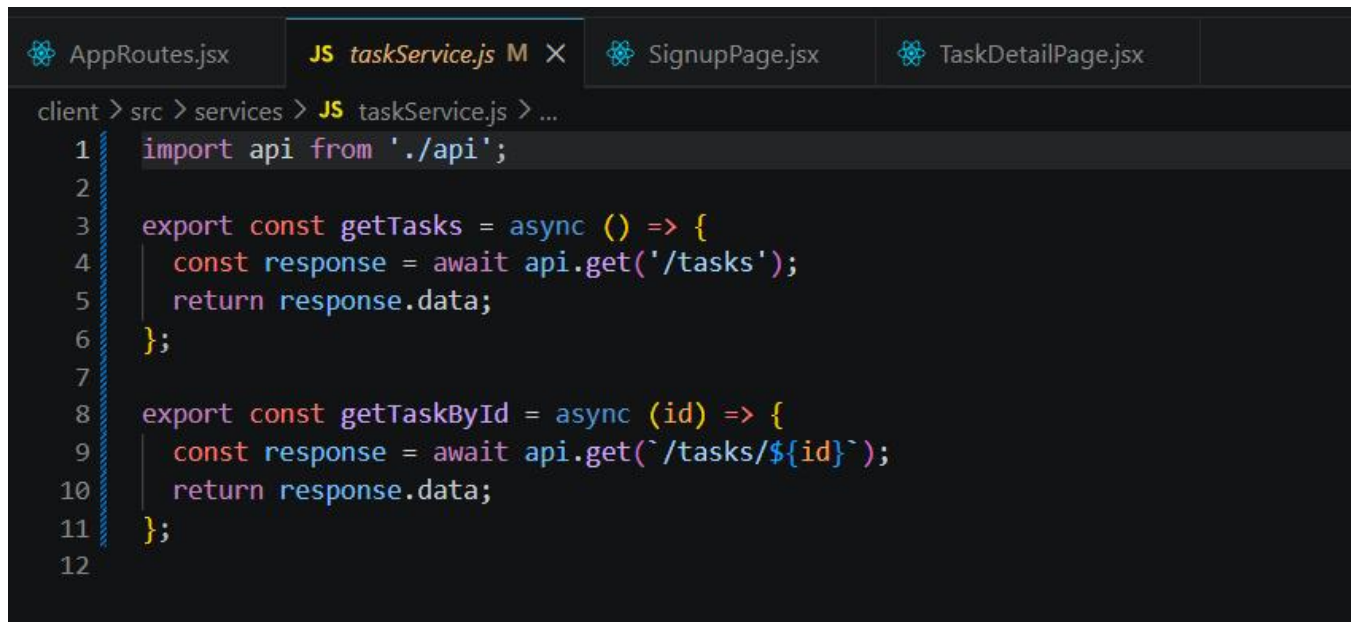
Figure 2.1 — Team service implementation for retrieving team data.

Task Service Implementation

- Implemented task service methods for task-related data retrieval.

Configured methods:

- Retrieve all tasks
 - Retrieve individual task details
- Integrated task service with the existing Axios configuration and authentication system.



```
client > src > services > JS taskService.js > ...
1  import api from './api';
2
3  export const getTasks = async () => {
4    const response = await api.get('/tasks');
5    return response.data;
6  };
7
8  export const getTaskById = async (id) => {
9    const response = await api.get(`/tasks/${id}`);
10   return response.data;
11 };
12
```

Figure 2.2 — Task service implementation for task data retrieval.

Dashboard Development

- Developed a dashboard page utilizing real backend data.

Dashboard components:

- Total Teams summary card
 - Total Tasks summary card
 - Recent Tasks section
- Configured concurrent API requests to improve dashboard loading performance.

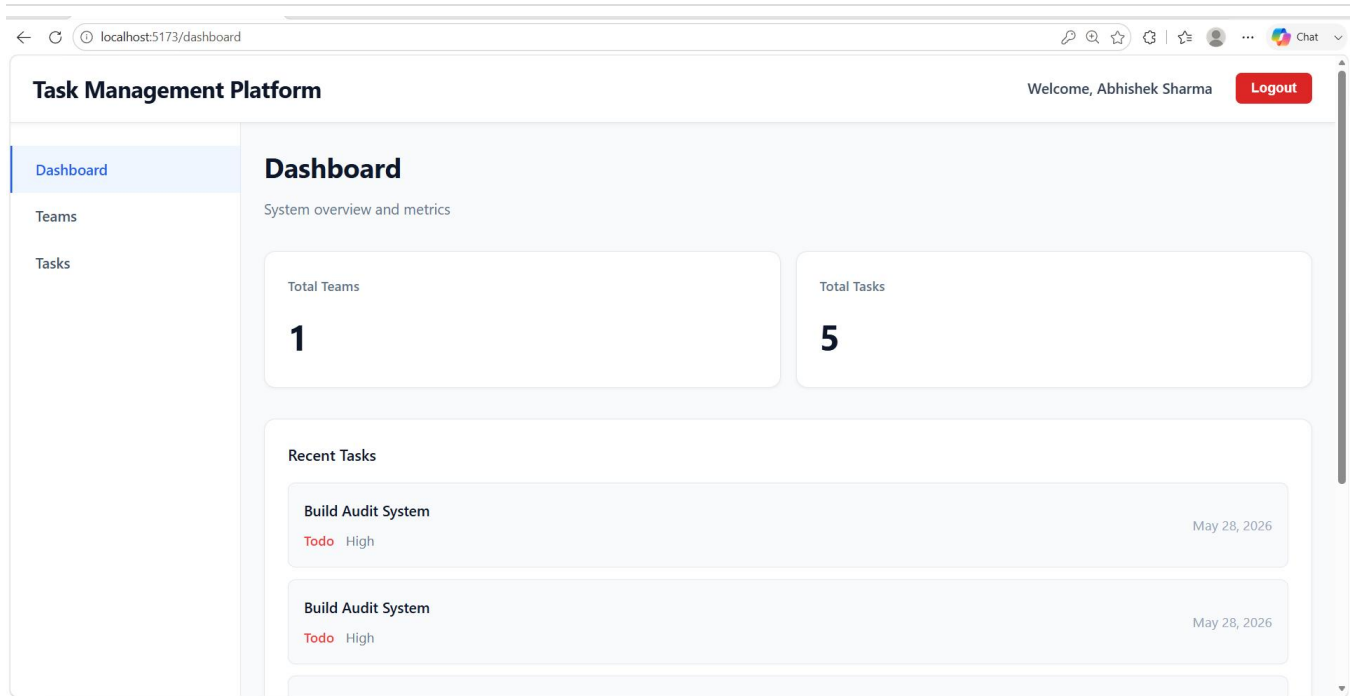


Figure 2.3 — Dashboard displaying team and task summary information.

Recent Task Integration

- Integrated task information into the dashboard interface.

Displayed information:

- Task title
 - Task status
 - Task priority
- Configured sorting and limited the display to the most recent tasks.

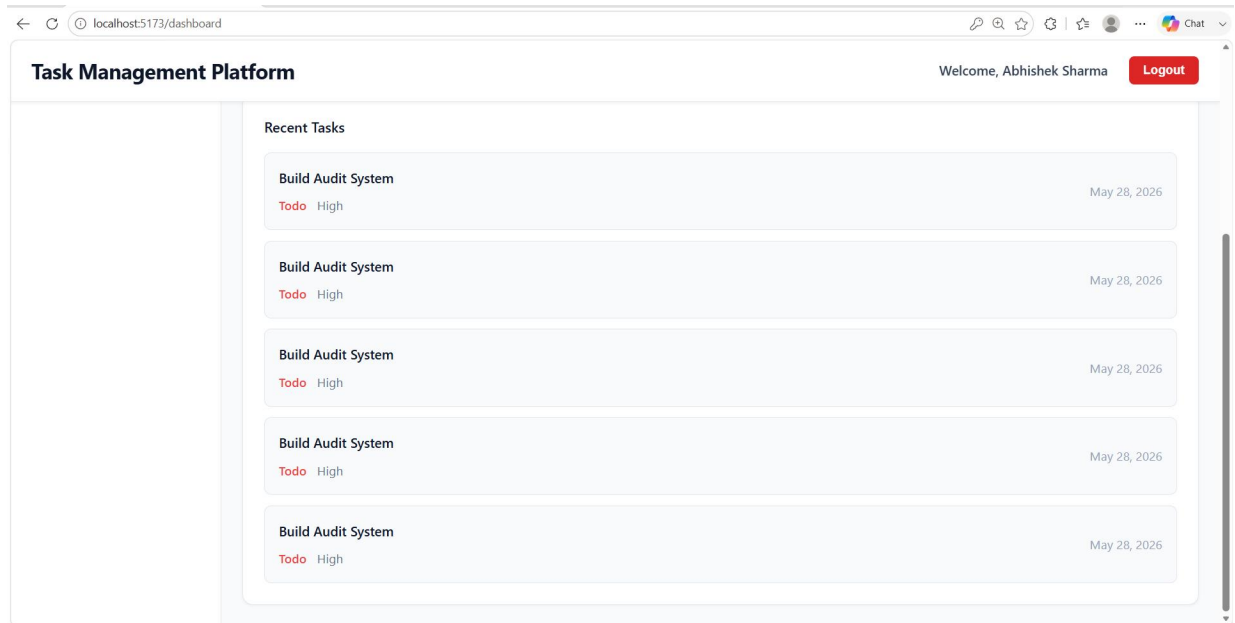


Figure 2.4 — Recent tasks retrieved from the backend and displayed on the dashboard.

Teams Page Development

- Implemented a team management overview page.

Displayed information:

- Team name
 - Member count
 - Creation date
- Presented team information using a card-based layout for improved readability.

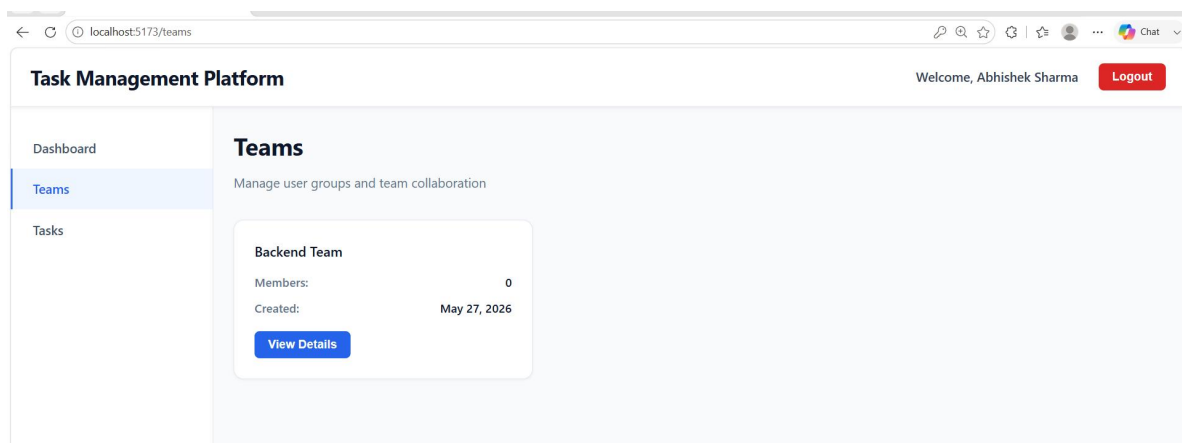


Figure 2.5 — Teams page displaying available teams and summary information.

Team Navigation Implementation

- Added navigation from team cards to dynamic team detail routes.

Configured functionality:

- Dynamic route generation
 - Team-specific navigation
 - React Router integration
- Verified route parameter handling through detail page navigation.

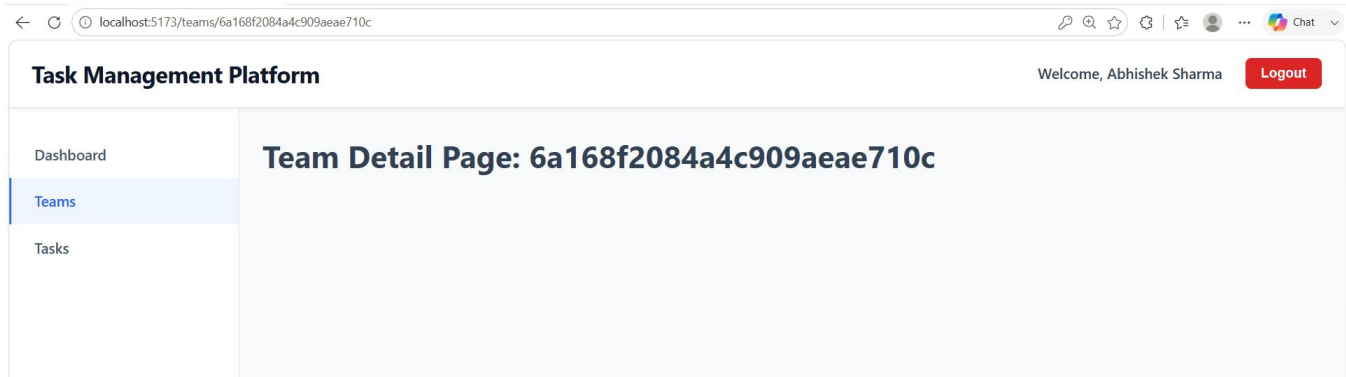


Figure 2.6 — Navigation to team detail page using dynamic route parameters.

Loading and Error Handling

- Implemented loading and error state management for asynchronous API operations.

Configured functionality:

- Dashboard loading state
 - Teams page loading state
 - Dashboard error state
 - Teams page error state
- Improved user experience during network requests and failure scenarios.

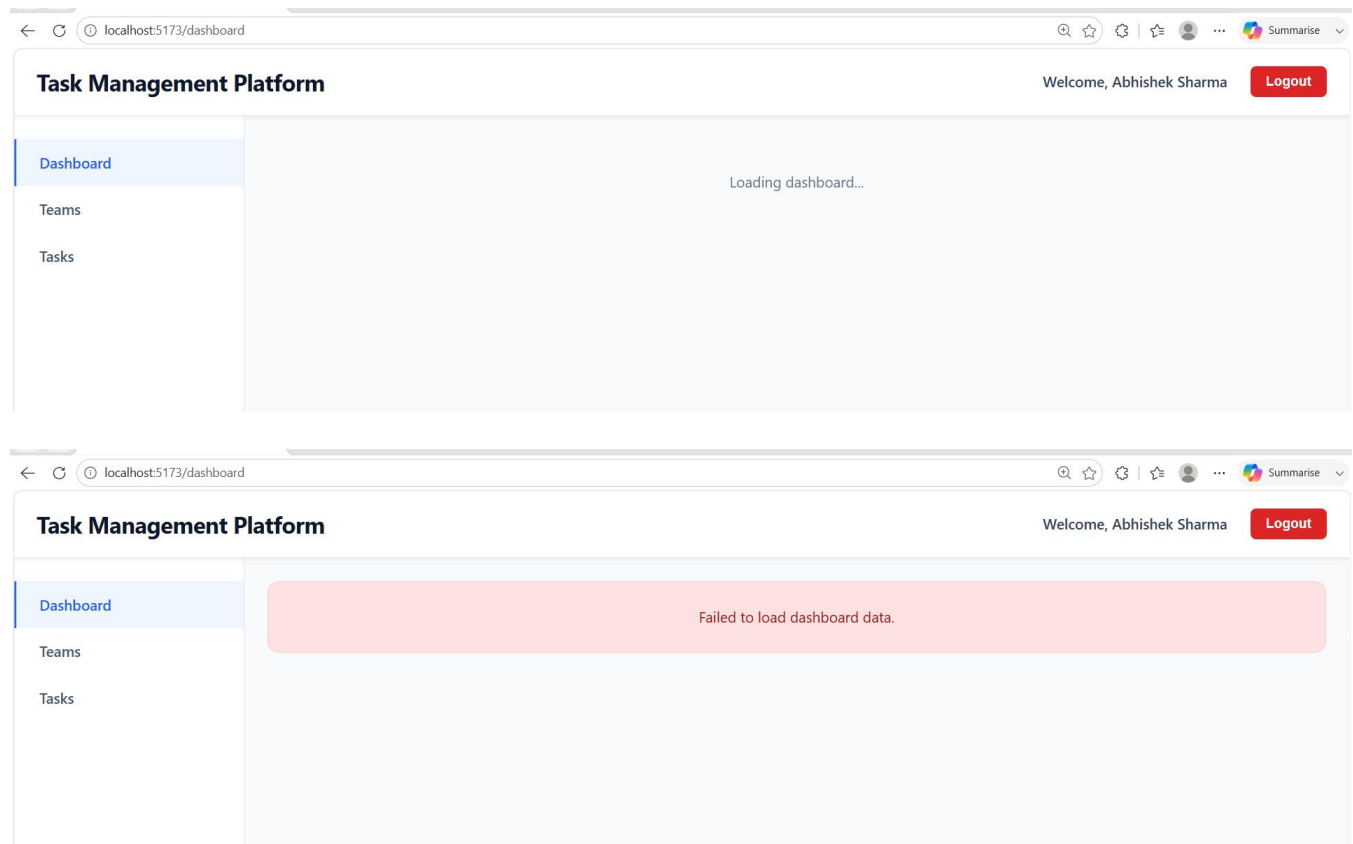


Figure 2.7 — Dashboard loading and error state handling implementation.

3. Conclusion

Day 17 focused on integrating backend data into the frontend interface and expanding application functionality through dashboard and team management features. Team and task services were successfully implemented, real-time data was displayed on the dashboard, team information was integrated into dedicated management pages, and loading and error handling mechanisms were established. The application now provides meaningful operational data to authenticated users and is prepared for detailed task and team management features in subsequent development phases.